CFD Assignment 3

Ambuj Nayan
24M0003
Aerospace Engineering
Indian Institute of Technology Bombay

I. Problem Statement

The objective of this study is to numerically determine the streamline pattern within a two-dimensional chamber (Figure 1) using the streamfunction equation:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0. \tag{1}$$

This represents steady, incompressible, and inviscid flow, with chamber walls acting as streamlines.

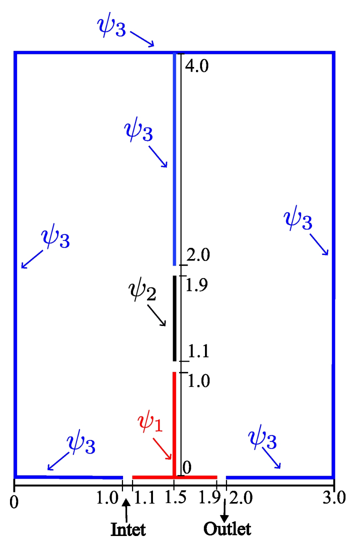


Figure 1: Domain of Problem Statement

A. Domain and Boundary Conditions

The chamber has an inlet, an outlet, and slots, each 0.1 m wide, with dimensions of 3 m (length) and 4 m (height). The assigned boundary conditions are:

	ψ_1	ψ_2	ψ_3
Test 1	100	150	300
Test 2	100	200	300
Test 3	100	250	300

Boundary conditions for different test cases

The computational grid is uniformly spaced with $\Delta x = \Delta y = 0.1$.

B. Numerical Method and Convergence Criteria

The solution is considered converged when the relative error, defined using the L2 norm, satisfies:

$$\text{ERROR} = \frac{\|\psi_{ij}^{n+1} - \psi_{ij}^{n}\|_2}{\|\psi_{ij}^{n+1}\|_2} < 10^{-4}, \tag{2}$$

where the L2 norm is given by:

$$\|\psi_{ij}\|_2 = \sqrt{\sum_{i=2}^{IM-1} \sum_{j=2}^{JM-1} |\psi_{ij}|^2}. \tag{3}$$

C. Effect of Initial Guess

To assess the influence of initial conditions on convergence, computations are performed for each test case with initial values of 100, 150, and 200.

II. Point Jacobi Iteration Method

The Jacobi iteration updates $u_{i,j}$ at iteration $k+1$ as [1]:

$$u_{i,j}^{k+1} = \frac{1}{2(1+\beta^2)} \left[u_{i+1,j}^{k} + u_{i-1,j}^{k} + \beta^2 \left(u_{i,j+1}^{k} + u_{i,j-1}^{k} \right) \right], \tag{4}$$

where k represents previous iteration values or initial guesses. The process continues until convergence.

III. Implementation

A. Symmetry

A symmetry line at $x = 1.5$ allows solving only half the domain ($0 \leq x \leq 1.5$ or $1.5 \leq x \leq 3.0$), reducing computational cost without affecting accuracy.

For a half-domain solution, the L2 norm is modified as:

$$\|\psi_{ij}\|_2 = \sqrt{\sum_{i=2}^{(IM-1)/2} \sum_{j=2}^{JM-1} |\psi_{ij}|^2}. \tag{5}$$

Despite this modification, the numerator and denominator in the relative error equation (equation 2) are both halved. As a result, the computed error remains unchanged, ensuring that the convergence criteria are unaffected by the reduced computational domain.

By solving only half of the domain and then mirroring the solution by the symmetry axis, we reduce the number of computational grid points required, thereby optimizing memory usage and computational efficiency. This approach simplifies implementation while yielding the same results as solving the full domain.

B. Flowchart

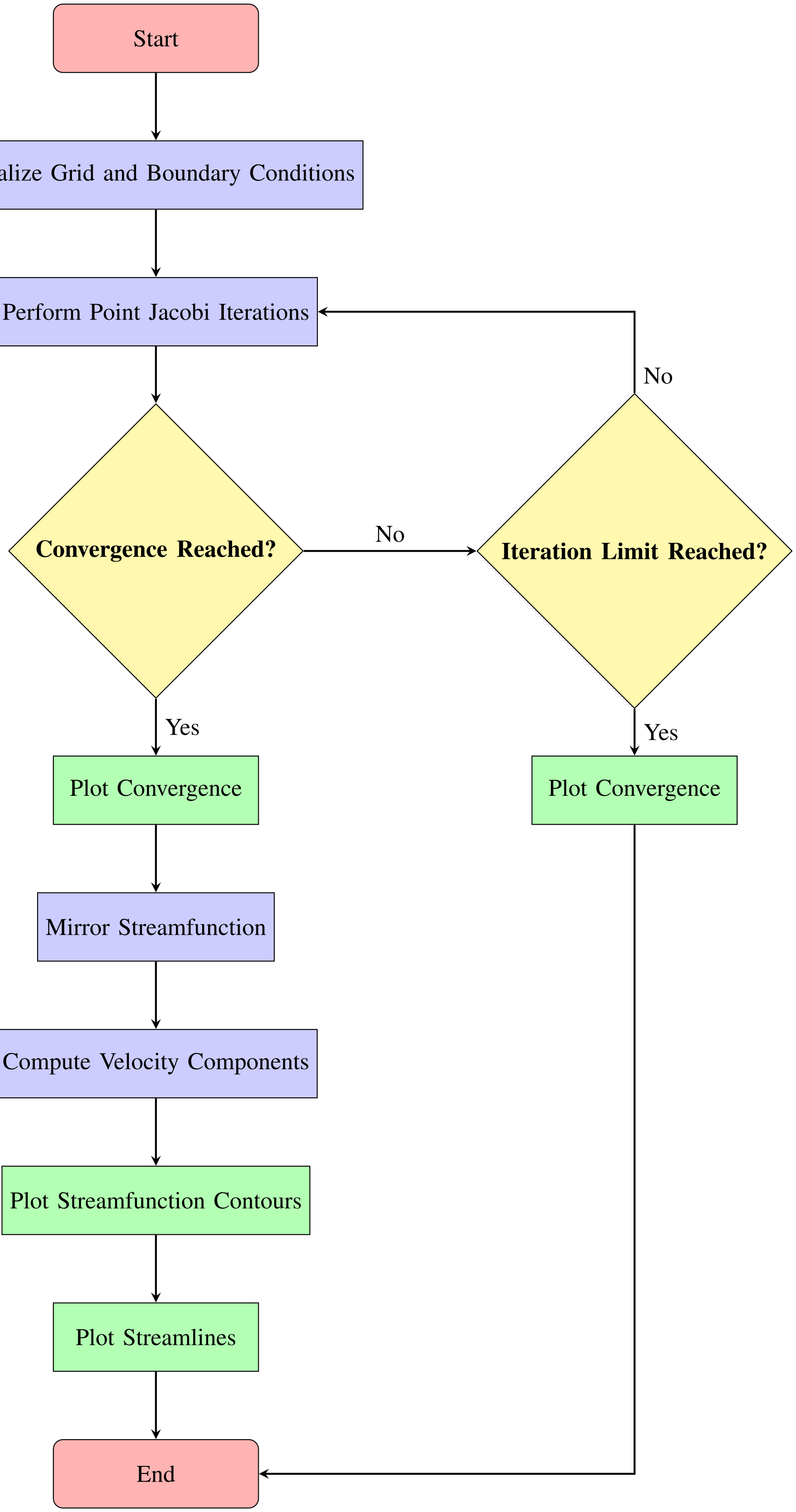


Figure 2: Flowchart for the Python Code

IV. Result

A. Test 1

$$\boldsymbol{\psi_1 = 100 \quad \psi_2 = 150 \quad \psi_3 = 300}$$

$\psi_{initial} = 100$

Test 1, Initial Guess 1				
y	**ψ(x=0)**	**ψ(x=1)**	**ψ(x=2)**	**ψ(x=3)**
0	300	300	300	300
0.1	300	232.8787	232.8787	300
0.2	300	211.2982	211.2982	300
0.3	300	201.7218	201.7218	300
0.4	300	196.2215	196.2215	300
0.5	300	192.7007	192.7007	300
0.6	300	190.5352	190.5352	300
0.7	300	189.5259	189.5259	300
0.8	300	189.6094	189.6094	300
0.9	300	190.7489	190.7489	300
1	300	192.8589	192.8589	300
1.1	300	195.7463	195.7463	300
1.2	300	199.1381	199.1381	300
1.3	300	202.9044	202.9044	300
1.4	300	207.0896	207.0896	300
1.5	300	211.8593	211.8593	300
1.6	300	217.4374	217.4374	300
1.7	300	224.0409	224.0409	300
1.8	300	231.778	231.778	300
1.9	300	240.4957	240.4957	300
2	300	249.6187	249.6187	300
2.1	300	258.2368	258.2368	300
2.2	300	265.7633	265.7633	300
2.3	300	272.023	272.023	300
2.4	300	277.0917	277.0917	300
2.5	300	281.1491	281.1491	300
2.6	300	284.3905	284.3905	300
2.7	300	286.9919	286.9919	300
2.8	300	289.0972	289.0972	300
2.9	300	290.8214	290.8214	300
3	300	292.2524	292.2524	300
3.1	300	293.4593	293.4593	300
3.2	300	294.494	294.494	300
3.3	300	295.3978	295.3978	300
3.4	300	296.2015	296.2015	300
3.5	300	296.9301	296.9301	300
3.6	300	297.6024	297.6024	300
3.7	300	298.2341	298.2341	300
3.8	300	298.8374	298.8374	300
3.9	300	299.423	299.423	300
4	300	300	300	300

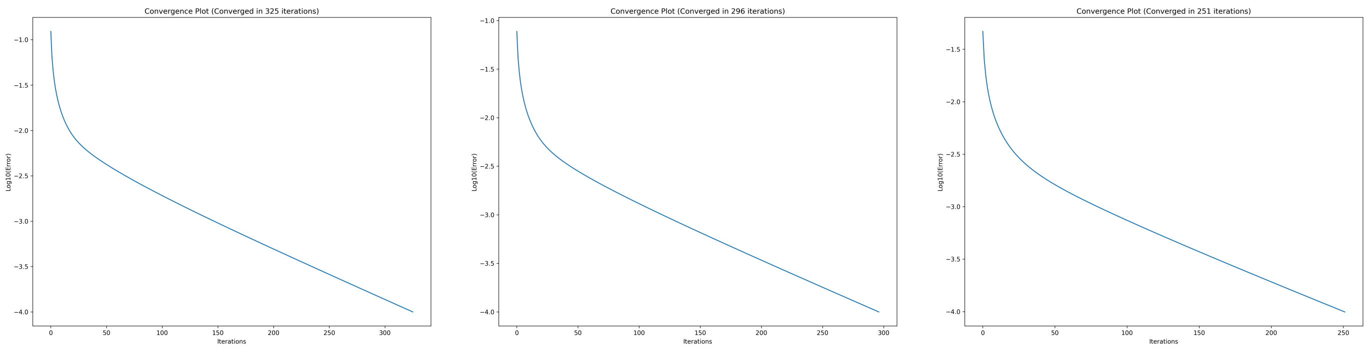
$\psi_{initial} = 150$

Test 1, Initial Guess 2			
ψ(x=0)	**ψ(x=1)**	**ψ(x=2)**	**ψ(x=3)**
300	300	300	300
300	232.8944	232.8944	300
300	211.3292	211.3292	300
300	201.7676	201.7676	300
300	196.2811	196.2811	300
300	192.7731	192.7731	300
300	190.6189	190.6189	300
300	189.6192	189.6192	300
300	189.7104	189.7104	300
300	190.8556	190.8556	300
300	192.9692	192.9692	300
300	195.8577	195.8577	300
300	199.2483	199.2483	300
300	203.0109	203.0109	300
300	207.1904	207.1904	300
300	211.9516	211.9516	300
300	217.5197	217.5197	300
300	224.1107	224.1107	300
300	231.8344	231.8344	300
300	240.5365	240.5365	300
300	249.6438	249.6438	300
300	258.2447	258.2447	300
300	265.7548	265.7548	300
300	271.9974	271.9974	300
300	277.0506	277.0506	300
300	281.0928	281.0928	300
300	284.3213	284.3213	300
300	286.9109	286.9109	300
300	289.0076	289.0076	300
300	290.7248	290.7248	300
300	292.1524	292.1524	300
300	293.3579	293.3579	300
300	294.395	294.395	300
300	295.3033	295.3033	300
300	296.1149	296.1149	300
300	296.8533	296.8533	300
300	297.5382	297.5382	300
300	298.1841	298.1841	300
300	298.8033	298.8033	300
300	299.4057	299.4057	300
300	300	300	300

$\psi_{initial} = 200$

Test 1, Initial Guess 3			
ψ(x=0)	**ψ(x=1)**	**ψ(x=2)**	**ψ(x=3)**
300	300	300	300
300	232.9459	232.9459	300
300	211.4312	211.4312	300
300	201.918	201.918	300
300	196.4772	196.4772	300
300	193.0108	193.0108	300
300	190.8937	190.8937	300
300	189.9255	189.9255	300
300	190.0424	190.0424	300
300	191.2064	191.2064	300
300	193.3318	193.3318	300
300	196.2249	196.2249	300
300	199.6125	199.6125	300
300	203.3651	203.3651	300
300	207.5267	207.5267	300
300	212.2639	212.2639	300
300	217.8009	217.8009	300
300	224.356	224.356	300
300	232.0378	232.0378	300
300	240.6955	240.6955	300
300	249.754	249.754	300
300	258.3061	258.3061	300
300	265.7652	265.7652	300
300	271.9594	271.9594	300
300	276.9644	276.9644	300
300	280.9634	280.9634	300
300	284.1515	284.1515	300
300	286.7079	286.7079	300
300	288.7761	288.7761	300
300	290.4736	290.4736	300
300	291.8877	291.8877	300
300	293.0894	293.0894	300
300	294.1295	294.1295	300
300	295.0502	295.0502	300
300	295.8809	295.8809	300
300	296.6466	296.6466	300
300	297.3643	297.3643	300
300	298.0489	298.0489	300
300	298.7107	298.7107	300
300	299.3587	299.3587	300
300	300	300	300

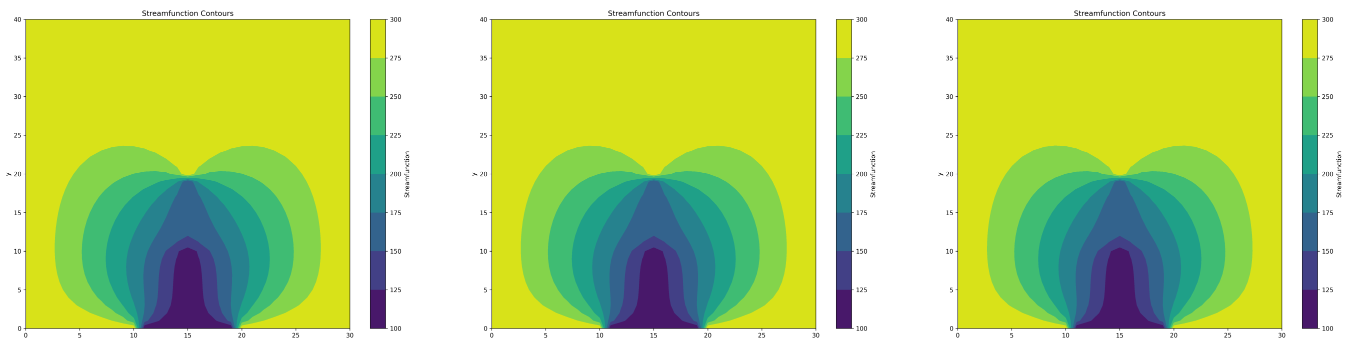
Table I: ψ values for 1st Test case with different Initial Values



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 3: Convergence Plots for different Initial Values of 1st Test Case

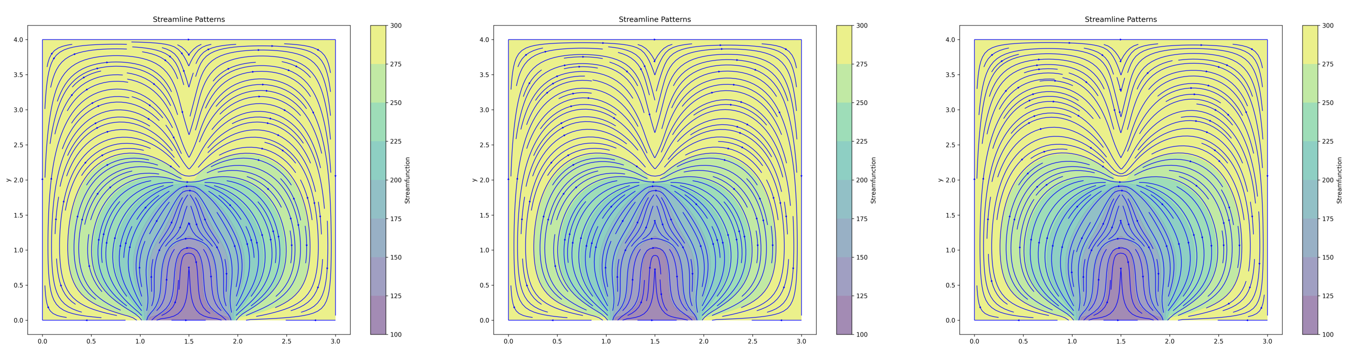
The third initial guess converged the fastest, requiring only 251 iterations, followed by the second initial guess at 296 iterations. The first initial guess took the longest, converging in 325 iterations. As shown in Table I, the third initial guess was closest to the final values, which explains its faster convergence, followed by the second and then the first.



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 4: Streamfunction Contour Plots for different Initial Values of 1st Test Case

Regardless of the initial guess, the results converge to nearly the same values, as evident from the above contour plot and Table I.



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 5: Streamline Plots for different Initial Values of 1st Test Case

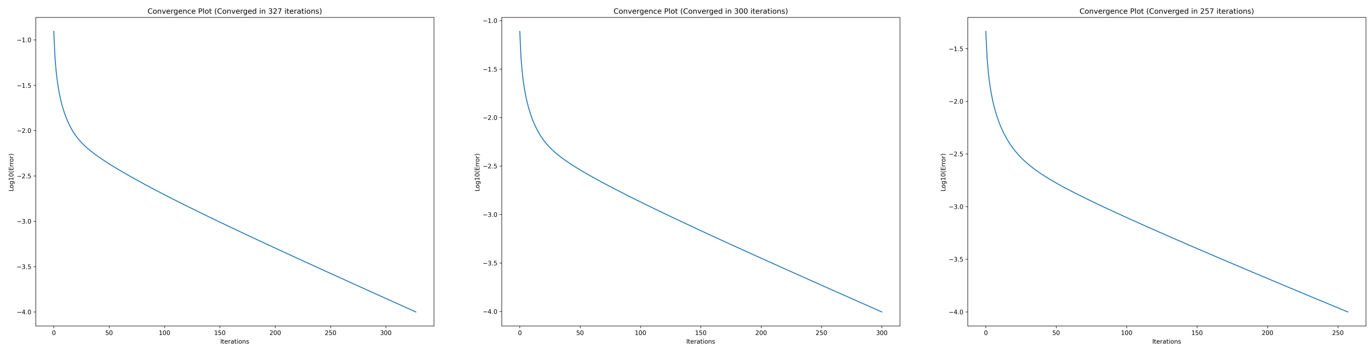
B. Test 2

$$\boldsymbol{\psi_1 = 100 \quad \psi_2 = 200 \quad \psi_3 = 300}$$

$\psi_{initial} = 100$ $\quad$ $\psi_{initial} = 150$ $\quad$ $\psi_{initial} = 200$

Test 2, Initial Guess 1					**Test 2, Initial Guess 2**				**Test 2, Initial Guess 3**			
y	**ψ(x=0)**	**ψ(x=1)**	**ψ(x=2)**	**ψ(x=3)**	**ψ(x=0)**	**ψ(x=1)**	**ψ(x=2)**	**ψ(x=3)**	**ψ(x=0)**	**ψ(x=1)**	**ψ(x=2)**	**ψ(x=3)**
0	300	300	300	300	300	300	300	300	300	300	300	300
0.1	300	233.4847	233.4847	300	300	233.4994	233.4994	300	300	233.535	233.535	300
0.2	300	212.5454	212.5454	300	300	212.5744	212.5744	300	300	212.6452	212.6452	300
0.3	300	203.6833	203.6833	300	300	203.7264	203.7264	300	300	203.8303	203.8303	300
0.4	300	199.0131	199.0131	300	300	199.0692	199.0692	300	300	199.2046	199.2046	300
0.5	300	196.4873	196.4873	300	300	196.5558	196.5558	300	300	196.7189	196.7189	300
0.6	300	195.5385	195.5385	300	300	195.6177	195.6177	300	300	195.806	195.806	300
0.7	300	196.027	196.027	300	300	196.1161	196.1161	300	300	196.3243	196.3243	300
0.8	300	197.9405	197.9405	300	300	198.0372	198.0372	300	300	198.262	198.262	300
0.9	300	201.2536	201.2536	300	300	201.3569	201.3569	300	300	201.5919	201.5919	300
1	300	205.8038	205.8038	300	300	205.9112	205.9112	300	300	206.1524	206.1524	300
1.1	300	211.176	211.176	300	300	211.2862	211.2862	300	300	211.5269	211.5269	300
1.2	300	216.76	216.76	300	300	216.8703	216.8703	300	300	217.1062	217.1062	300
1.3	300	222.1885	222.1885	300	300	222.2975	222.2975	300	300	222.522	222.522	300
1.4	300	227.3975	227.3975	300	300	227.5027	227.5027	300	300	227.7116	227.7116	300
1.5	300	232.5181	232.5181	300	300	232.6181	232.6181	300	300	232.8055	232.8055	300
1.6	300	237.7695	237.7695	300	300	237.8619	237.8619	300	300	238.0241	238.0241	300
1.7	300	243.3742	243.3742	300	300	243.4579	243.4579	300	300	243.5902	243.5902	300
1.8	300	249.4764	249.4764	300	300	249.5496	249.5496	300	300	249.6492	249.6492	300
1.9	300	256.0322	256.0322	300	300	256.0939	256.0939	300	300	256.1578	256.1578	300
2	300	262.705	262.705	300	300	262.7541	262.7541	300	300	262.7808	262.7808	300
2.1	300	268.9237	268.9237	300	300	268.9597	268.9597	300	300	268.9482	268.9482	300
2.2	300	274.3251	274.3251	300	300	274.3477	274.3477	300	300	274.2981	274.2981	300
2.3	300	278.8133	278.8133	300	300	278.8224	278.8224	300	300	278.7358	278.7358	300
2.4	300	282.4555	282.4555	300	300	282.4517	282.4517	300	300	282.3298	282.3298	300
2.5	300	285.3839	285.3839	300	300	285.3677	285.3677	300	300	285.2137	285.2137	300
2.6	300	287.7386	287.7386	300	300	287.7112	287.7112	300	300	287.5286	287.5286	300
2.7	300	289.6444	289.6444	300	300	289.6069	289.6069	300	300	289.4007	289.4007	300
2.8	300	291.2033	291.2033	300	300	291.1576	291.1576	300	300	290.9325	290.9325	300
2.9	300	292.4963	292.4963	300	300	292.4439	292.4439	300	300	292.2062	292.2062	300
3	300	293.5857	293.5857	300	300	293.529	293.529	300	300	293.2844	293.2844	300
3.1	300	294.5202	294.5202	300	300	294.4608	294.4608	300	300	294.2165	294.2165	300
3.2	300	295.3365	295.3365	300	300	295.277	295.277	300	300	295.0389	295.0389	300
3.3	300	296.0634	296.0634	300	300	296.0056	296.0056	300	300	295.7807	295.7807	300
3.4	300	296.7228	296.7228	300	300	296.669	296.669	300	300	296.4629	296.4629	300
3.5	300	297.3318	297.3318	300	300	297.2836	297.2836	300	300	297.1025	297.1025	300
3.6	300	297.9035	297.9035	300	300	297.8629	297.8629	300	300	297.7114	297.7114	300
3.7	300	298.4486	298.4486	300	300	298.4167	298.4167	300	300	298.2993	298.2993	300
3.8	300	298.9752	298.9752	300	300	298.9534	298.9534	300	300	298.8732	298.8732	300
3.9	300	299.4904	299.4904	300	300	299.4792	299.4792	300	300	299.4386	299.4386	300
4	300	300	300	300	300	300	300	300	300	300	300	300

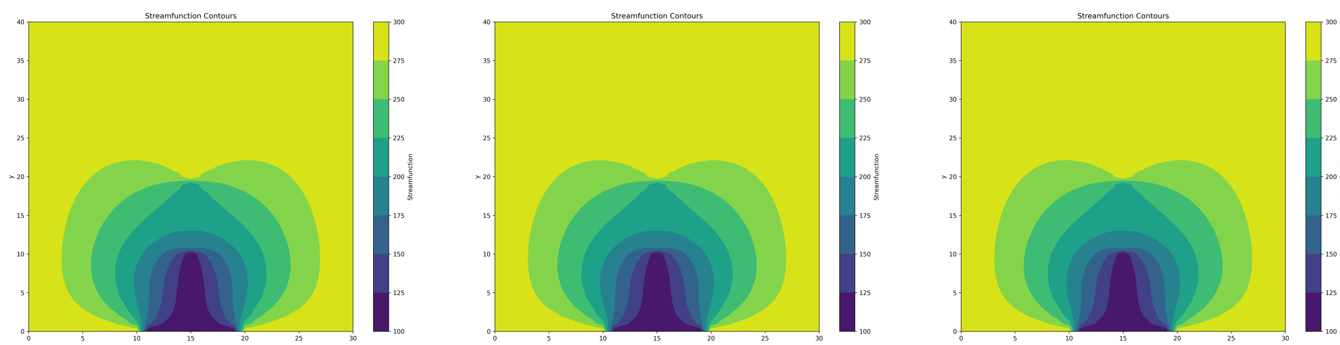
Table II: ψ values for 2nd Test case with different Initial Values



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 6: Convergence Plots for different Initial Values of 2nd Test Case

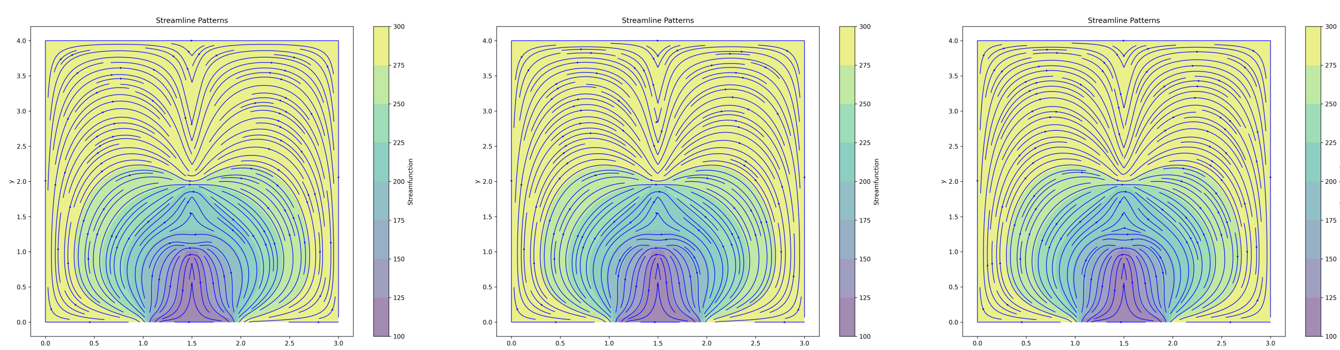
As seen in previous case here also the third initial guess converged the fastest, requiring only 257 iterations, followed by the second initial guess at 300 iterations. The first initial guess took the longest, converging in 327 iterations. As shown in Table II, the third initial guess was closest to the final values, which explains its faster convergence, followed by the second and then the first.



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 7: Streamfunction Contour Plots for different Initial Values of 2nd Test Case

Similar to the previous test case, the results in this case also converge to nearly the same values regardless of the initial guess, as evident from the above contour plot and Table II.



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 8: Streamline Plots for different Initial Values of 2nd Test Case

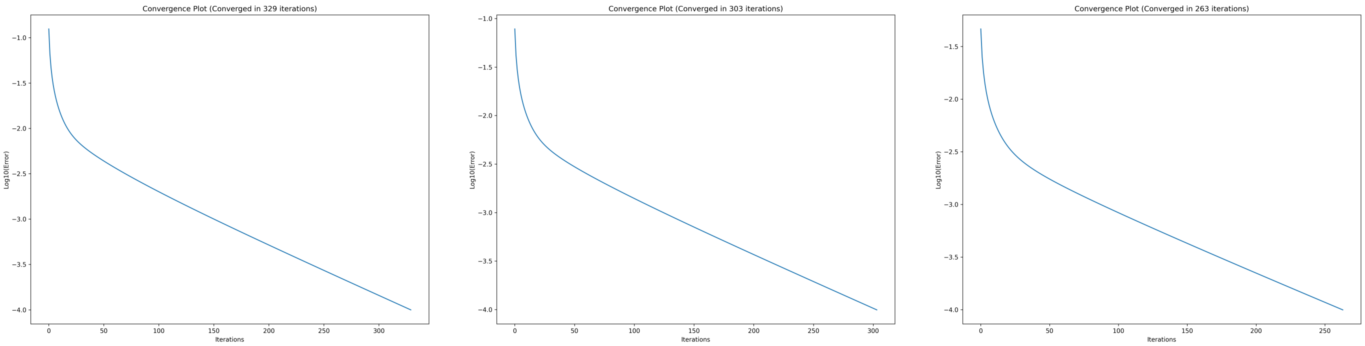
C. Test 3

$$\boldsymbol{\psi_1 = 100 \quad \psi_2 = 250 \quad \psi_3 = 300}$$

$\psi_{initial} = 100$ $\psi_{initial} = 150$ $\psi_{initial} = 200$

Test 3, Initial Guess 1					Test 3, Initial Guess 2				Test 3, Initial Guess 3			
y	ψ(x=0)	ψ(x=1)	ψ(x=2)	ψ(x=3)	ψ(x=0)	ψ(x=1)	ψ(x=2)	ψ(x=3)	ψ(x=0)	ψ(x=1)	ψ(x=2)	ψ(x=3)
0	300	300	300	300	300	300	300	300	300	300	300	300
0.1	300	234.0912	234.0912	300	300	234.1025	234.1025	300	300	234.1294	234.1294	300
0.2	300	213.7939	213.7939	300	300	213.8163	213.8163	300	300	213.8694	213.8694	300
0.3	300	205.6467	205.6467	300	300	205.6797	205.6797	300	300	205.7578	205.7578	300
0.4	300	201.8072	201.8072	300	300	201.8504	201.8504	300	300	201.952	201.952	300
0.5	300	200.2769	200.2769	300	300	200.3293	200.3293	300	300	200.4517	200.4517	300
0.6	300	200.5454	200.5454	300	300	200.6062	200.6062	300	300	200.7472	200.7472	300
0.7	300	202.5322	202.5322	300	300	202.6002	202.6002	300	300	202.7559	202.7559	300
0.8	300	206.2761	206.2761	300	300	206.3502	206.3502	300	300	206.5179	206.5179	300
0.9	300	211.7632	211.7632	300	300	211.8418	211.8418	300	300	212.0169	212.0169	300
1	300	218.7539	218.7539	300	300	218.8358	218.8358	300	300	219.0149	219.0149	300
1.1	300	226.6113	226.6113	300	300	226.6947	226.6947	300	300	226.873	226.873	300
1.2	300	234.3878	234.3878	300	300	234.4713	234.4713	300	300	234.6455	234.6455	300
1.3	300	241.4785	241.4785	300	300	241.5603	241.5603	300	300	241.7256	241.7256	300
1.4	300	247.7116	247.7116	300	300	247.7904	247.7904	300	300	247.9435	247.9435	300
1.5	300	253.1832	253.1832	300	300	253.2573	253.2573	300	300	253.3938	253.3938	300
1.6	300	258.1078	258.1078	300	300	258.176	258.176	300	300	258.2934	258.2934	300
1.7	300	262.7137	262.7137	300	300	262.7745	262.7745	300	300	262.8692	262.8692	300
1.8	300	267.1809	267.1809	300	300	267.2333	267.2333	300	300	267.3035	267.3035	300
1.9	300	271.5745	271.5745	300	300	271.6175	271.6175	300	300	271.6608	271.6608	300
2	300	275.7971	275.7971	300	300	275.8301	275.8301	300	300	275.8458	275.8458	300
2.1	300	279.616	279.616	300	300	279.6385	279.6385	300	300	279.6256	279.6256	300
2.2	300	282.8921	282.8921	300	300	282.904	282.904	300	300	282.863	282.863	300
2.3	300	285.6084	285.6084	300	300	285.6096	285.6096	300	300	285.5411	285.5411	300
2.4	300	287.8239	287.8239	300	300	287.8148	287.8148	300	300	287.7206	287.7206	300
2.5	300	289.6229	289.6229	300	300	289.604	289.604	300	300	289.486	289.486	300
2.6	300	291.0906	291.0906	300	300	291.063	291.063	300	300	290.9242	290.9242	300
2.7	300	292.3004	292.3004	300	300	292.265	292.265	300	300	292.1088	292.1088	300
2.8	300	293.3126	293.3126	300	300	293.2708	293.2708	300	300	293.1012	293.1012	300
2.9	300	294.174	294.174	300	300	294.1272	294.1272	300	300	293.9485	293.9485	300
3	300	294.9216	294.9216	300	300	294.8715	294.8715	300	300	294.6883	294.6883	300
3.1	300	295.5832	295.5832	300	300	295.5316	295.5316	300	300	295.3486	295.3486	300
3.2	300	296.1808	296.1808	300	300	296.1293	296.1293	300	300	295.9514	295.9514	300
3.3	300	296.7306	296.7306	300	300	296.681	296.681	300	300	296.513	296.513	300
3.4	300	297.2454	297.2454	300	300	297.1993	297.1993	300	300	297.0457	297.0457	300
3.5	300	297.7346	297.7346	300	300	297.6936	297.6936	300	300	297.5585	297.5585	300
3.6	300	298.2055	298.2055	300	300	298.1709	298.1709	300	300	298.0581	298.0581	300
3.7	300	298.6637	298.6637	300	300	298.6367	298.6367	300	300	298.5493	298.5493	300
3.8	300	299.1134	299.1134	300	300	299.0949	299.0949	300	300	299.0352	299.0352	300
3.9	300	299.5579	299.5579	300	300	299.5485	299.5485	300	300	299.5183	299.5183	300
4	300	300	300	300	300	300	300	300	300	300	300	300

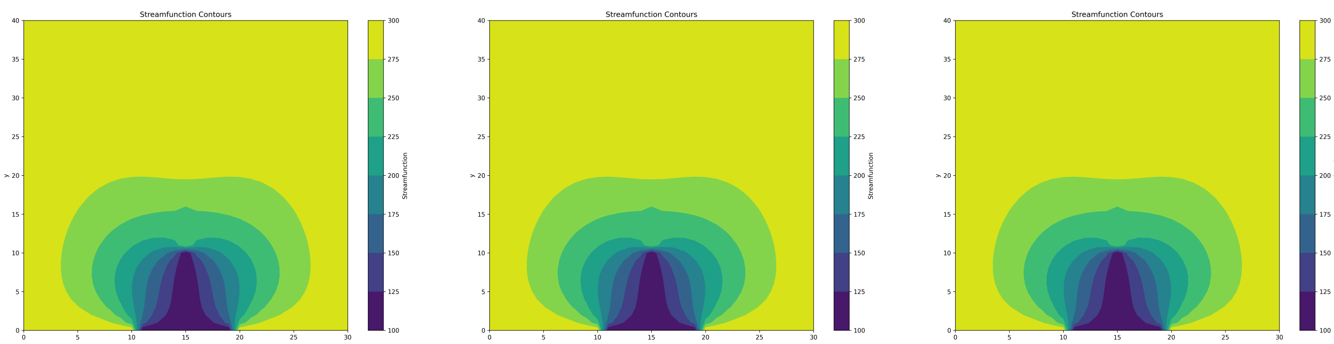
Table III: ψ values for 3rd Test case with different Initial Values



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 9: Convergence Plots for different Initial Values of 3rd Test Case

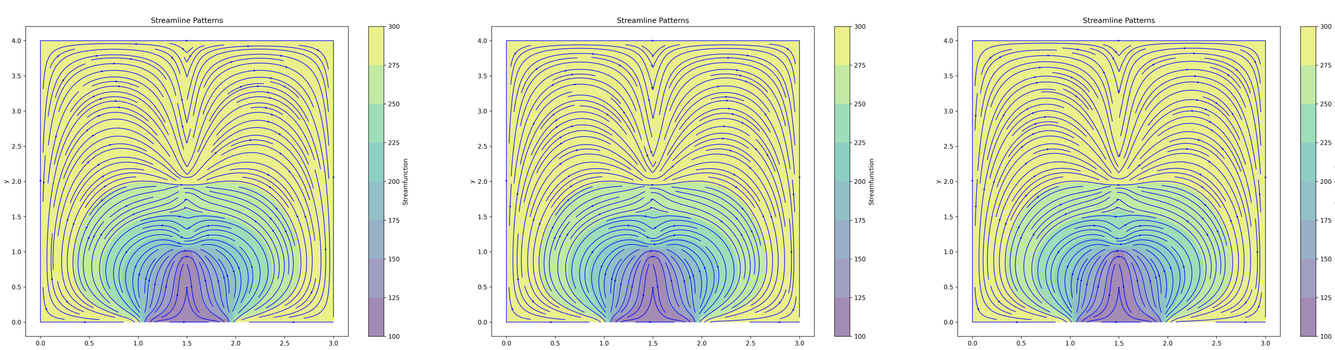
As seen in previous case here also the third initial guess converged the fastest, requiring only 263 iterations, followed by the second initial guess at 303 iterations. The first initial guess took the longest, converging in 329 iterations. As shown in Table III, the third initial guess was closest to the final values, which explains its faster convergence, followed by the second and then the first.



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 10: Streamfunction Contour Plots for different Initial Values of 3rd Test Case

Similar to the previous test case, the results in this case also converge to nearly the same values regardless of the initial guess, as evident from the above contour plot and Table III.



(a) $\psi_{initial} = 100$ (b) $\psi_{initial} = 150$ (c) $\psi_{initial} = 300$

Figure 11: Streamline Plots for different Initial Values of 3rd Test Case

Appendix

References

[1] T. J. Chung, *Computational Fluid Dynamics*. Cambridge University Press, 2010.